CAPSTONE · AI FOR GOOD · WEEK 6

The supply chain *and China*

Europe wants its wind to be cheap and its energy to be secure. Tonight: why, in the supply chain, those two goals pull in opposite directions.

Dr. Jan Erik Meidell · Wednesday 25 November 2026 · online · 18:15–20:00 · keep the shared canvas open on a second screen

TONIGHT

One hour forty-five, two exercises

18:18	A guess in the chat
18:21	Who makes the world's turbines, and at what price
18:32	Western turbine makers in trouble
18:38	Below the turbine: components, steel, rare earths, ships, cables
18:55	Exercise 1 · your turbine's dependency map
19:12	Break · 5 minutes
19:17	Chinese makers at Europe's door, and doors closing
19:28	Europe's answer: rules, and their price
19:40	Exercise 2 · advise the Italian government
19:53	Into Phase 2 · close

A GUESS IN THE CHAT · A PERCENTAGE

Of all the wind capacity connected worldwide in 2025, how much was in China?

About 73%: 120.5 GW of about 165 GW. In 2024 it was about 68% (79.8 of 117 GW).

China is not only the largest maker of turbines. It is also by far the largest buyer, and that home market is what lets its manufacturers scale.

GWEC, Global Wind Report 2025 and 14 May 2026 release; the percentages are computed.

WHO MAKES THE TURBINES · 2025

The world's top five are all Chinese

TURBINE MAKER	GW INSTALLED, 2025
Goldwind	**29.7**
Envision	**21.8**
Windey	**19.8**
Mingyang	**18.6**
SANY	**15.1**
Vestas, the leader outside China	***12.9***

But 93.4% of what Chinese makers installed was in China. In Europe they installed 446 MW, less than one large offshore farm.

GWEC, 14 May 2026. Manufacturing capacity, 2023: China 60% of 163 GW, Europe 19%, US 9% (GWEC via Reuters).

THE PRICE GAP

A Chinese turbine costs 20–50% less

About 20%

cheaper than US and European turbines, on average (BloombergNEF, 2024)

Up to 50%

cheaper "in some cases", with up to three years of deferred payment (WindEurope, May 2024)

+45%

rise in Western turbine prices since 2020, to US$1.0–1.2 m per MW (Wood Mackenzie, Nov 2025)

Why so cheap? Scale at home, integrated supply chains, cheaper steel and labour, and, the EU suspects, state support. The EU is investigating that last point.

BY HAND · WHAT THE PRICE GAP IS WORTH

The turbine is a third of the farm

STEP	PER MW
Week 4's farm: capex €4,770k, turbines 35% of it	**€1,670k**
A turbine 20% cheaper saves	**€334k = 7% of capex**
A turbine 50% cheaper saves	**€835k = 17.5% of capex**
Effect on the cost of each MWh (from €114, at 6.2%)	**–€5.3 to –€13.3 (–5% to –12%)**

Real money, but less than two points on the cost of capital (+16%). Cheap turbines are not the only way to cheap wind.

The Week 4 calculator: capex down by 7% and by 17.5%, everything else unchanged.

WESTERN TURBINE MAKERS

Selling below cost

SIEMENS GAMESA	FY2023	FY2024	FY2025
Profit before special items	**–€4,347 m**	**–€1,781 m**	**–€1,364 m**
Margin	**–47.8%**	**–17.8%**	**–13.1%**

What went wrong

Blade and bearing problems on new onshore models (about €1.6 bn of charges in 2023), offshore ramp-up costs, and fixed-price orders signed before costs rose.

The others

Vestas: back to a 5.7% margin in 2025. GE Vernova: blade failures at Dogger Bank, where Eni's Plenitude holds a stake, and at Vineyard Wind in 2024.

Siemens Energy earnings releases; Vestas annual report 2025; company and press reports on GE Vernova.

WEEK 4'S LESSON, ONE LEVEL DOWN

The squeeze travels down the chain

Auctions pushed prices down → developers pushed turbine prices down → turbine makers rushed bigger models to market below cost → quality problems and losses → fewer, more expensive Western suppliers.

A balancing loop for your report §5: a lower auction price weakens the supply chain that delivers the next auction.

BELOW THE TURBINE

China's share of the parts

Castings 82% · converters 82%

The heavy iron hubs and frames; the power electronics that connect the turbine to the grid

Gearboxes 80% · generators 73%

The drivetrain inside the nacelle

Steel 53%

of the world's crude steel in 2024: 1,005 of 1,885 Mt. Towers and foundations are mostly steel.

Bearings and blades

No reliable recent share found. Say so in your report rather than guessing.

GWEC Global Wind Report 2024 (component shares, probably 2023 data); worldsteel provisional 2024 data, share computed.

RARE EARTHS

The magnets in offshore turbines

~60%

mining

China's share of magnet rare-earth mining, 2024

91%

refining

China's share of refining them

94%

magnets

China's share of sintered permanent magnets (about 50% in 2005)

Large offshore turbines often use direct-drive generators with permanent magnets. BloombergNEF estimates offshore turbines use almost four times as much rare earth per MW as onshore.

IEA, rare earth elements, 2025; BNEF via Reuters, April 2025.

WHEN SUPPLY BECOMES POLICY

China's rare-earth export controls

DATE	WHAT HAPPENED
4 April 2025	Export licences required for seven rare earths and for magnets. Some carmakers in the US and EU cut production; European rare-earth prices reached up to six times Chinese prices.
9 October 2025	Controls extended to five more elements, and to foreign-made products containing more than 0.1% Chinese rare earths
After the US–China deal	The October package suspended until **10 November 2026**; the April controls stay

That suspension ran out yesterday. What happened? Look it up in the next minute and type it in the chat.

IEA commentary, 23 October 2025.

SHIPS AND CABLES

Who can install it, and who can connect it

Installation vessels, to 2030

China's fleet can install about **225 GW**. Outside China about **175 GW**, if €2.5–4 bn a year goes into new vessels for three years, and about **140 GW** if it does not. 500 GW is "out of reach".

Kühne Climate Center, January 2026: vessel capacity, not a forecast.

Subsea power cables

Three firms, NKT, **Prysmian (Italian)** and Nexans, hold about 75% of the high-voltage market, booked into the late 2020s. Europe may be short by up to 27% a year in cables above 132 kV, about 22,800 km over 2026–2040.

FT via Offshore Magazine; TGS | 4C, June 2026.

SORTING · TYPE C OR E

Where does Europe depend most on China?

1. The magnets in a direct-drive offshore turbine. **China: 94% of sintered magnets**
2. High-voltage subsea cable. **Europe: NKT, Prysmian, Nexans**
3. Castings and converters. **China: about 82% each**
4. The offshore turbines installed in Europe so far. **Europe: Siemens Gamesa, Vestas**

Europe still builds most of its own offshore turbines. The dependency sits one or two levels below them.

EXERCISE 1 · 13 MINUTES · BREAKOUT ROOMS

Your turbine's dependency map

1. Take one 15 MW offshore turbine for a farm in your team's **most ambitious** 2040 world.
2. In the canvas table **"Dependency map"**, one row each: turbine · magnets · castings · steel · installation vessel · export cable. For each: **who supplies it today**, and **what happens to your world** if that supply stops for a year.
3. Circle the one row that would break your world first.

Back in the main room at 19:08. One person per team names its breaking row.

READ-OUT

Which row breaks your world?

Most teams pick magnets or vessels. Note which: it is a bottleneck for report §4, with the deployment level at which it binds.

BREAK

Five minutes

Back at 19:17. Cameras on when you return.

ITALY · CHINESE TURBINES AT SEA

The first Chinese turbines in European waters

Beleolico, Taranto, 2022

30 MW, ten Mingyang 3 MW turbines, built by Renexia (Toto group). The first offshore wind farm in the Mediterranean, and the first in Europe with Chinese turbines.

What came next

August 2024: Mingyang, Renexia and the Italian government sign a memorandum for a turbine factory in Italy. October 2024: Mingyang wins the design contract for **Med Wind**, a 2.8 GW floating farm off Sicily with 18.8 MW turbines.

offshorewind.biz, 2022 and 2024. Whether a factory site has been chosen is not confirmed; check before you cite it.

AND DOORS CLOSING

Security enters the purchase decision

WHERE	WHAT HAPPENED
Germany · Waterkant, 300 MW	Mingyang named preferred supplier in July 2024; in August 2025 the developer switched to Siemens Gamesa after the Defence Ministry raised security concerns
United Kingdom · Ardersier, Scotland	A proposed £1.5 bn Mingyang factory with about 1,500 jobs; blocked on national-security grounds in March 2026
European Union	A review of Chinese turbine suppliers under the Foreign Subsidies Regulation from April 2024; an in-depth investigation into Goldwind from February 2026, with 18 months to conclude

A modern turbine is connected, monitored and controlled remotely. Buying one is also a decision about who has access to the grid.

THE DILEMMA

Cheap or secure?

Buy Chinese

Turbines 20–50% cheaper, financing offered, faster delivery. Cheaper wind, sooner, at a lower cost to consumers and taxpayers.

Buy European

Jobs and factories at home, no foreign access to control systems, less exposure to export controls. Paid for in a higher price per MWh.

There is no right answer, only a price on security. Your report should say what that price is in your worlds.

EUROPE'S ANSWER

Rules that put a price on resilience

RULE	WHAT IT DOES
Net-Zero Industry Act (2024)	A goal: EU factories cover at least 40% of the EU's needs for clean-energy technology by 2030. From 30 December 2025, auctions must use non-price criteria (cybersecurity, sustainability, resilience) for 30% of volumes, or 6 GW a year per country.
Critical Raw Materials Act (2024)	By 2030: 10% of EU needs mined in the EU, 40% processed, 25% recycled, and no more than 65% from any single country
Wind Power Package (October 2023)	15 actions to keep the industry European, as wind grows from 204 GW (2022) to more than 500 GW by 2030

YOUR TWO COMPANIES

Who is exposed to what?

Ørsted

Buys large offshore turbines from European makers. Exposed to their quality problems, their price rises and their capacity, and through magnets to Chinese supply further down. A resilience rule in auctions helps its suppliers, and raises its costs.

Eni / Plenitude

A minority stake in Dogger Bank, where GE's blades failed. A home market, Italy, that has already used Chinese turbines at sea and has an Italian cable champion, Prysmian. Spread across solar and onshore wind, with different supply chains.

FOR YOUR SCENARIOS

Supply-chain openness is a good axis

Open ↔ weaponised supply chains: a genuine driver, and largely independent of EU climate policy. A strong-policy Europe can still face export controls; a weak-policy Europe can still import cheap turbines.

If your team's axes failed the Week 4 test, this is a candidate replacement. If they passed, it belongs in your worlds as a constraint on each deployment number.

EXERCISE 2 · 11 MINUTES · BREAKOUT ROOMS

Advise the Italian government

Med Wind, 2.8 GW off Sicily. Should Italy allow Chinese turbines in it?

1. Decide: **yes**, **no**, or **yes, if…**
2. Give the three strongest arguments, one of them with a number from tonight.
3. If "yes, if": name the one condition, for example on control systems, local assembly or data.

In the canvas box "Advice to Rome". Back at 19:51; the other team gets one question.

READ-OUT

Two teams, one decision

Did you put a price on security? A good answer says how many euros per MWh, or how many years of delay, Italy should accept for it.

THIS WEEK · THE PHASE 2 PROPOSAL IS DUE FRIDAY

Where tonight goes

TONIGHT	WHERE
The breaking row of your dependency map	**§4 · The bottlenecks**
Supply-chain openness as an axis or a constraint	**§6 · The four worlds**
A supply-chain indicator: rare-earth prices, the Chinese share of European turbine orders, vessels on order	**§8 · The Phase 2 proposal**: question, indicators, method, risks

CLOSE

Cheap wind and secure wind *are not the same purchase*

Europe will pay for its wind either in euros or in dependence. A good consulting report says which, and how much.

Questions on the material: the forum of the week. Anything personal: the message panel, bottom right of every page.